

A MATRIX METHODOLOGY FOR SPECTRAL ANALYSIS OF WAVE PROPAGATION IN MULTIPLE CONNECTED TIMOSHENKO BEAMS

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The dynamics of the Timoshenko beam is recast such that the description requires information only at the end points. This is then presented in the form of a dynamic stiffness relation suitable for assembling, as is done analogously for conventional finite elements. The analysis gives the exact frequency dependent response for the Timoshenko beam irrespective of the length of the element. The frequency domain response is converted to the time domain response using the Fast Fourier Algorithm (FFT). Examples of its use in wave propagation studies are given.

1. INTRODUCTION

A previous paper by Farris and Doyle [1] separated the dynamics of a split beam into a *global* analysis of a structure composed of uniform waveguides and a *local* analysis at the crack tip. By imposing the appropriate dynamic connectivities between the waveguides, a system of equations was set up that allowed determining the coefficients associated with the generated waves.

A drawback to the approach is that as the number of waveguides increases (as with a finite split in the beam, say), handling all the connectivities becomes unwieldy. This paper, therefore, introduces a matrix methodology for the convenient assembling of structures made of many Timoshenko beams. It has characteristics very similar to that of the Finite Element Method, but it is formulated in the frequency domain and the basic “element” is a complete waveguide, extending from one structural joint to another. This paper is a continuation of the work presented in references [2, 3] in which spectrally formulated elements for rods and beams, respectively, are presented. The significant difference here is that a spectral element for the higher order Timoshenko beam theory needs to be developed. Indeed, the primary scope of the following analysis is the development of such an element.

Conventional finite elements treat the dynamic load induced by the mass and rotational inertia of the beam as concentrated loads and moments applied at the ends of the element. Even though the structural joints may be far apart, many elements must be used if the inertia distributions are to be modeled accurately. Therefore, the number of elements required to do a dynamic problem is substantially larger than that required for the equivalent static one. This paper applies the methods of spectral analysis (see reference [4]) to formulate an element which treats the distribution of mass and rotational inertia exactly. Only one spectral element need be placed between any two joints, substantially reducing the total number of degrees of freedom in the system. The spectral formulation requires that the assembled system of equations be solved in the frequency domain and utilizes the

Fast Fourier Transform (FFT) to convert the frequency domain results back to the time domain.

The formulation develops along the same line as that for the elementary beam as done in reference [3]. First, the spectral relations for the Timoshenko beam are discussed. Then the spectral element is developed and its reduction to the elementary Bernoulli–Euler beam is shown in the limit as the frequency tends to zero. Two other theories are also recovered when the shear stiffness GAK tends to infinity and the rotational inertia ρI tends to zero, respectively. (There is no such problem as element or shear “locking” as occurs in conventional formulations. According to reference [5], the origin of this problem lies in the order of the interpolating functions, and the spectral approach essentially uses the exact interpolation function.) A “throw-off” (or one noded) element is also introduced to handle those situations with semi-infinite members. Examples of impact-induced wave propagation in various connected Timoshenko beams are given.

2. SPECTRAL ANALYSIS OF THE TIMOSHENKO BEAM

The relevant equations of motion for the Timoshenko beam have already appeared in reference [1], but for completeness a brief summary is given here. With the introduction of shear deformation, the assumption of the elementary theory that plane sections remain plane is no longer valid. This implies that we cannot obtain the slope $\phi(x, t)$ of any section along the length of the beam simply by differentiation of the transverse displacement $v(x, t)$. Thus, we have two independent motions $v(x, t)$ and $\phi(x, t)$ and the equations of motion for these are

$$\begin{aligned} GAK \left[\frac{\partial^2 v}{\partial x^2} - \frac{\partial \phi}{\partial x} \right] &= \rho A \frac{\partial^2 v}{\partial t^2}, \\ EI \frac{\partial^2 \phi}{\partial x^2} + GAK \left[\frac{\partial v}{\partial x} - \phi \right] &= \rho I \frac{\partial^2 \phi}{\partial t^2}. \end{aligned} \quad (1)$$

The shear and bending stiffnesses are governed by GAK and EI , respectively, while the corresponding inertias are ρA and ρI . The waveguide resultants of shear force and bending moments acting on any section are related to v and ϕ by

$$V = GAK \left[\frac{\partial v}{\partial x} - \phi \right] = -EI \frac{\partial^2 \phi}{\partial x^2} + \rho I \frac{\partial^2 \phi}{\partial t^2}, \quad (2)$$

$$M = EI \frac{\partial \phi}{\partial x}. \quad (3)$$

Spectral analysis represents the solutions in the form

$$v(x, t) = \sum \hat{v} e^{-i(kx - \omega t)}, \quad \phi(x, t) = \sum \hat{\phi} e^{-i(kx - \omega t)}, \quad (4)$$

where k is the wavenumber, ω is the angular frequency and $i = \sqrt{-1}$. The summations are over frequency and the amplitude spectrums $\hat{v}$ and $\hat{\phi}$ are frequency dependent. These solutions, on substitution into the equations of motions, give four possible modes as:

$$k(\omega) = \pm \left[\frac{1}{2} \left\{ \left(\frac{1}{c_s} \right)^2 + \left(\frac{Q}{c_0 q} \right)^2 \right\} \omega^2 \pm \sqrt{\left(\frac{\omega}{c_0 q} \right)^2 + \frac{1}{4} \left\{ \left(\frac{1}{c_s} \right)^2 - \left(\frac{Q}{c_0 q} \right)^2 \right\} \omega^4} \right]^{1/2}, \quad (5)$$

where the constants $c_0 q \equiv \sqrt{EI/\rho A}$, $c_s \equiv \sqrt{GAK/\rho A}$ and $Q \equiv \sqrt{\rho I/\rho A}$ have been introduced. It also gives that the deflections and rotations are related by

$$\hat{v} = i \left[\frac{EI k^2 + GAK - \rho I \omega^2}{GAKk} \right] \hat{\phi} \quad (6)$$

for each mode. Consequently, it is sufficient to take either $\hat{v}$ or $\hat{\phi}$ as the basic unknown. Typically, the complete solution is given as

$$\begin{aligned} \hat{\phi}(x) &= \mathbf{A} e^{-ik_1 x} + \mathbf{B} e^{-ik_2 x} + \mathbf{C} e^{+ik_1 x} + \mathbf{D} e^{+ik_2 x}, \\ \hat{v}(x) &= P_1 \mathbf{A} e^{-ik_1 x} + P_2 \mathbf{B} e^{-ik_2 x} - P_1 \mathbf{C} e^{+ik_1 x} - P_2 \mathbf{D} e^{+ik_2 x}, \end{aligned} \quad (7)$$

where k_1 and k_2 are the wavenumbers of mode 1 and mode 2, respectively, and $\{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$ are the undetermined coefficients dependent on frequency. P_1 and P_2 are the amplitude ratios given by equation (6) with the appropriate value of k substituted.

The Bernoulli-Euler beam relations can be recovered by setting, separately, $Q=0$ (no rotational inertia) and $GAK=\infty$ (no shear deformation). The higher order theory for beams does not introduce additional modes as compared to the elementary theory. However, the behavior of each mode is changed, as can be seen from Figure 1. In the limit, as the frequency becomes very small, then,

$$k_1 \simeq \sqrt{\omega/(c_0 q)}, \quad k_2 \simeq i \sqrt{\omega/(c_0 q)}. \quad (8)$$

These are the Bernoulli-Euler spectrum relations as discussed in reference [4]. In the limit, as the frequency becomes very large, then

$$k_1 \Rightarrow \omega/c_s, \quad k_2 \Rightarrow \omega Q/c_0 q, \quad (9)$$

both of which, being real, are propagating solutions. The complete dispersion relations

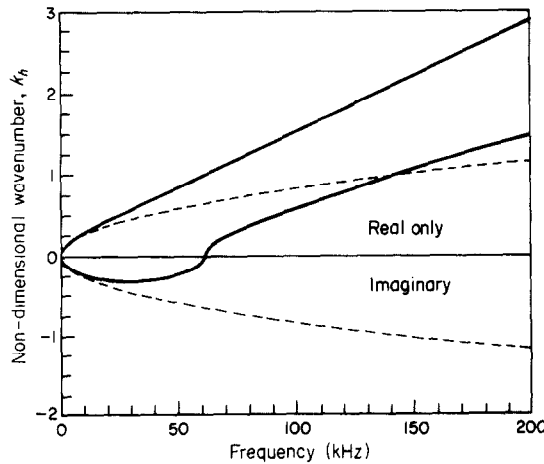


Figure 1. Spectrum relation for the Timoshenko beam, (—) and the Bernoulli-Euler beam (-----).

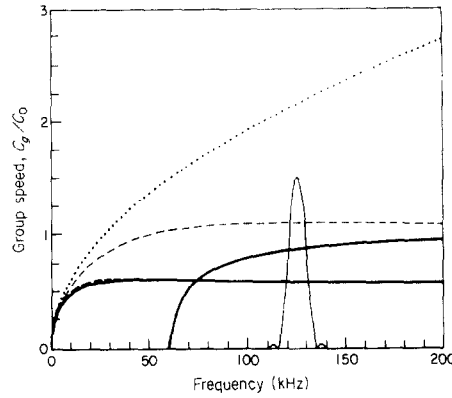


Figure 2. Dispersion relation for the Timoshenko beam where $c_0 \equiv \sqrt{EA/\rho A}$. $\cdots$, Bernoulli-Euler; —, Timoshenko; ---, Timoshenko (shear only); - · - ·, Rayleigh; —, pulse spectrum.

for the group speeds (defined as $c_g = d\omega/dk$) are shown in Figure 2. One of the significant differences between the elementary beam and the higher order beam theory is the presence of the second propagating mode. Therefore, in the examples to follow, this will be one of the features that are emphasized.

3. SPECTRAL ELEMENTS

Two special cases arise in the dynamics of connected beams. The first is when the member extends to infinity and is connected at only one point. This will be referred to as a single-noded or throw-off element. The other is when the beam is connected at both ends. The behavior of both of these are fundamentally different and so will be treated separately.

3.1. THROW-OFF OR SINGLE NODDED ELEMENT

This element behaves as a throw-off bar in that it is a conduit for energy out of the system. Unlike a typical absorber, it can absorb all frequency components. The stiffness relation for the element is established by taking the shape function as

$$\hat{\phi}(x) = \mathbf{A} e^{-ik_1 x} + \mathbf{B} e^{-ik_2 x}, \quad \hat{v}(x) = P_1 \mathbf{A} e^{-ik_1 x} + P_2 \mathbf{B} e^{-ik_2 x}. \quad (10)$$

Since this is a single-noded element, we consider only the degrees of freedom at node 1, as shown in Figure 3(a), and its corresponding nodal loads. These shape functions can be written in terms of modal displacement $v_1 \equiv v(0)$ and rotation $\phi_1 \equiv \phi(0)$, thus establishing

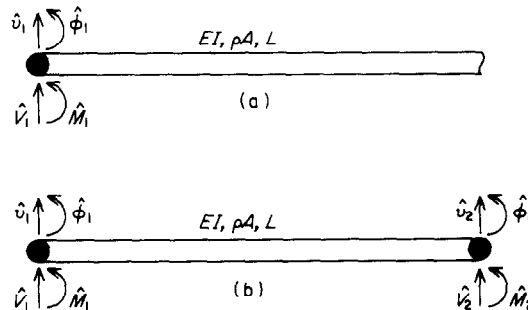


Figure 3. Nodal loads and degrees of freedom for element formulation.

a relation between the coefficients $\{\mathbf{A}, \mathbf{B}\}$ and the nodal degrees of freedom. The resulting equations can be written as

$$\begin{Bmatrix} \mathbf{A} \\ \mathbf{B} \end{Bmatrix} = [\hat{Q}] \{\hat{u}\} = \frac{1}{P_1 - P_2} \begin{bmatrix} 1 & -P_2 \\ -1 & P_1 \end{bmatrix} \begin{Bmatrix} \hat{v}_1 \\ \hat{\phi}_2 \end{Bmatrix}. \quad (11)$$

The nodal force and moment are obtained from the resultants equations (2) and (3), respectively. Then, using $V_1 \equiv -V(0)$ and $M_1 \equiv -M(0)$, a relation is established between the nodal forces and the coefficients which can be written as

$$\begin{Bmatrix} \hat{V}_1 \\ \hat{M}_1 \end{Bmatrix} = EI \begin{bmatrix} -(k_1^2 - \omega^2/c_r^2) & -(k_2^2 - \omega^2/c_r^2) \\ ik_1 & ik_2 \end{bmatrix} \begin{Bmatrix} \mathbf{A} \\ \mathbf{B} \end{Bmatrix} \quad (12)$$

and the constant $c_r = \sqrt{EI/\rho I}$. Substituting for the coefficients in terms of the nodal degrees of freedom, we arrive finally at the stiffness relation

$$\begin{Bmatrix} \hat{V}_1 \\ \hat{M}_1 \end{Bmatrix} = EI \begin{bmatrix} \hat{k}_{11} & \hat{k}_{12} \\ \hat{k}_{12} & \hat{k}_{22} \end{bmatrix} \begin{Bmatrix} \hat{v}_1 \\ \hat{\phi}_1 \end{Bmatrix}. \quad (13)$$

The individual stiffness terms are

$$\hat{k}_{11} = \frac{k_2^2 - k_1^2}{P_1 - P_2}, \quad \hat{k}_{12} = \frac{i(k_1 - k_2)}{P_1 - P_2}, \quad \hat{k}_{22} = \frac{i(P_1 k_2 - P_2 k_1)}{P_1 - P_2}.$$

This throw-off stiffness matrix is complex and, therefore, the structural stiffness matrix will also be complex after assembling. Keep in mind that a complex stiffness will exhibit a damped behavior in the time domain. In actuality, what is happening is that the bar is conducting energy out of the system. This element is very useful when the time of interest is short and the structure is large.

3.2. FINITE LENGTH ELEMENT

Consider a waveguide of finite length L , with loads applied only at its ends as shown in Figure 3(b). We write the shape functions as

$$\begin{aligned} \hat{\phi}(x) &= \mathbf{A} e^{-ik_1 x} + \mathbf{B} e^{-ik_2 x} + \mathbf{C} e^{-ik_1(L-x)} + \mathbf{D} e^{-ik_2(L-x)}, \\ \hat{v}(x) &= P_1 \mathbf{A} e^{-ik_1 x} + P_2 \mathbf{B} e^{-ik_2 x} + \mathbf{C} e^{-ik_1(L-x)} - P_2 \mathbf{D} e^{-ik_2(L-x)}. \end{aligned} \quad (14)$$

These functions are the exact solutions to the governing differential equations and they cannot be viewed as interpolating functions as commonly used in the conventional formulations. In addition, there are only four unknown coefficients as apposed to a minimum of six that is normally used in finite elements. These functions can be written in terms of nodal displacements $v_1 \equiv v(0)$ and $v_2 \equiv v(L)$, and rotations $\phi_1 \equiv \phi(0)$ and $\phi_2 \equiv \phi(L)$, thus establishing a relation between the coefficients $\{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$ and the nodal degrees of freedom. Inverting the relation, the resulting equations can be written symbolically as

$$\{\mathbf{A}\} = [\hat{Q}] \{\hat{u}\}. \quad (15)$$

The explicit form for the 4×4 matrix $[\hat{Q}]$ is given in the Appendix. As before, the shear forces and moments are written in terms of their nodal values using the resultants relation of equations (2) and (3), respectively. Substituting these into the above, we obtain a

relation between the nodal loads and the degrees of freedom which can be written in the form

$$\{\hat{F}\} = \frac{EI}{L^3} [\hat{k}] \{\hat{u}\}, \quad (16)$$

where the nodal degree of freedom vector $\{\hat{u}\}$ has the components $\{\hat{v}_1, \hat{\phi}_1, \hat{v}_2, \hat{\phi}_2\}$, and the nodal force vector $\{\hat{F}\}$ has the components $\{\hat{V}_1, \hat{M}_1, \hat{V}_2, \hat{M}_2\}$. The individual stiffness terms of the 4×4 matrix $[\hat{k}]$ are also given in the Appendix.

The matrix $[\hat{k}]$ has the symmetry properties of the static stiffness matrix and in general, is complex. (When there is no damping, however, it is real only.) In Figure 4 is shown a comparison between the stiffness for the Timoshenko beam, the Bernoulli-Euler beam, and the conventional element. It is apparent that all have the same limiting behavior for

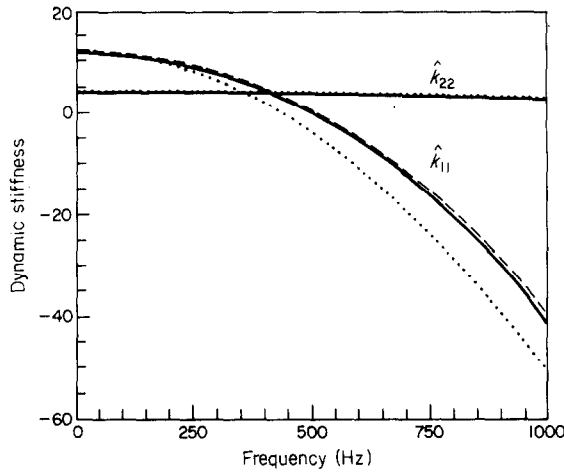


Figure 4. Dynamic stiffness at low frequencies. —, Timoshenko; ---, Bernoulli-Euler; ·····, conventional.

low frequency. Note that this occurs when the parameter $\xi = kL$ is small, i.e., low frequency or small element size. The behavior at high frequencies is quite different, as shown in Figure 5 for the representative stiffness $\hat{k}_{11}$. The conventional element continues to infinity and is not shown. The spectral elements, on the other hand, have multiple zeros with a shape similar to a frequency response curve. Note that the peaks show a definite drift as the frequency increases, with the peaks of the Timoshenko theory occurring at the lower frequencies.

4. ASSEMBLAGE

The spectral analysis approach to dynamic problems essentially gives rise to a sequence of pseudo-static problems. Hence, all the usual assemblage procedures from the conventional formulation are available. Indeed, there is a complete analog to the 3-D equations and reference is made to the papers by Richard and Leung [6] and Langley [7]. The structural system of equations can be presented as

$$[\hat{K}]\{\hat{u}\} = \{\hat{P}\}, \quad (17)$$

where, for the 2-D case, $\{\hat{P}\}$ is the applied load vector made up of applied forces $\hat{P}_x, \hat{P}_y$ and applied moments $\hat{T}_z$, at each node, and the vector $\{\hat{u}\}$ is composed of the nodal

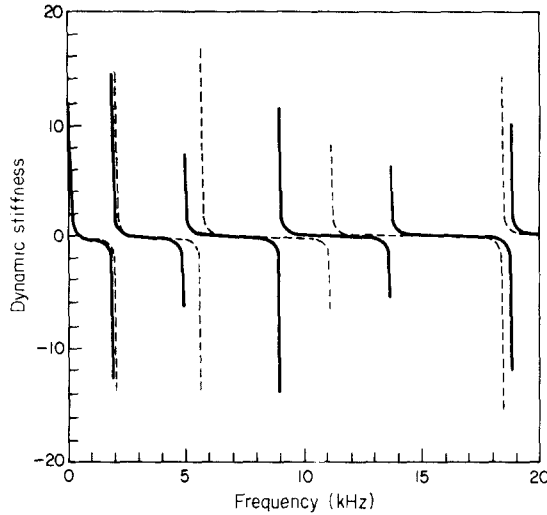


Figure 5. Dynamic stiffness $\hat{k}_{11}$ at high frequencies. —, Timoshenko; ---, Bernoulli-Euler.

degrees of freedom $\hat{u}$, $\hat{v}$ and $\hat{\phi}$, also at each node. The structural stiffness matrix $[\hat{K}]$ has all the usual properties including being symmetric and banded. Unlike the case of reference [6], however, in general the matrix is complex.

4.1. COMPUTER IMPLEMENTATION

The spectral formulation, as described so far, is implemented in a prototype program called SPECODYN [8]. It has been tested on both mainframe as well as on a microcomputer and the program is fully 3-D. The program structure is almost identical to that of a conventional finite element program in terms of input, system solving and output. However, the assemblage is done as a part of the *do-loop* over all the frequency components. For each member, the local stiffness is established and then transformed into global coordinates. This is entered into the structural stiffness matrix simply by associating the appropriate nodal numbers. After elimination of all zero degrees of freedom, the resulting structural stiffness matrix is stored in upper banded form. A $[U]^T[D][U]$ decomposition, followed by back substitution, is used to solve for the generalized nodal displacements. This solver must handle complex algebra as well as possible negative diagonal terms, but otherwise is the same as used for positive definite systems. The nodal displacement vector $\{\hat{u}\}$ is solved for at each frequency. Therefore, the element stiffness must be evaluated and then assembled at each frequency.

It is worth pointing out that the procedure for obtaining the spectral response is to first obtain the generalized displacements at all nodes and store them on the disk. (This is feasible because of the limited number of them.) Then, as part of the post-processing, the response at particular nodes or locations are obtained by reading the nodal displacements back in and doing the necessary computations. This process takes a negligible amount of time. As a final step, the time reconstructions are obtained by performing an inverse FFT.

4.2. OUTPUT FOR ARBITRARY LOCATION

Since the spectral element can be very large compared to the conventional element, the ability to compute the response between the nodes is necessary. This can be done by using the nodal displacement to first give the coefficients $[A, B, \dots]$ for the member, and then

using relations (2) and (3), combined with equation (7), to obtain the member distributions.

This ability is a significant boon to an experimentalist since the placement of sensors (strain gages or accelerometers, say) is not restricted by the element discretization. This is especially important in wave studies because small changes in position can give significant changes in response.

5. EXAMPLES

The aim of these examples is to demonstrate some of the capabilities of the spectral approach. The first example is used to illustrate the presence of the second propagating mode in the Timoshenko beam theory. To do this, the frequency spectrum of the input force is made narrow banded with the central frequency positioned beyond the cut-off frequency. For the present beam parameters, the cut-off frequency is at $\omega_0 = (GAK/\rho I)^{1/2} = 62$ kHz, where nominal aluminum properties are assumed as given in Table 1, and a cross-section of 25 mm $\times$ 6 mm is used. The second example explores the manner in which a spectrum response (which is the likely case in impact loading) is different for the Timoshenko and the Bernoulli-Euler beam models. The results are also compared with a conventional finite element solution. The third example shows that potentially useful quantities such as the system transfer function are automatic by-products of the approach. All examples are for simple structures to help focus ideas.

TABLE 1
Material properties and parameters used in the examples

Material Property	Value
E	70 GPa
G	27 GPa
ρ	2700 kg/m ³
K	0.85
Width	6 mm
Depth	25 mm

5.1. EXAMPLE (A): IMPACT OF AN INFINITE BEAM

The amplitude spectrum $\hat{F}$ for the input force is shown in Figure 2. This was obtained from a sinusoidally modulated pulse using a 2048-point FFT transform with a sampling rate ΔT of 0.5 μ s.

In this example, two throw-off elements are used to illustrate the presence of the second propagating mode. The arrangement is shown in Figure 6, with all elements having the same properties. The reduced stiffness matrix is 4×4 , with the unknown nodal degrees of freedom being $\{\hat{u}\} = \{\hat{v}_1, \hat{\phi}_1, \hat{v}_2, \hat{\phi}_2\}$. The rotation $\hat{\phi}_1$ could be imposed to be zero, but satisfaction of this condition is used as a check. The beam is impacted at node 1 and the response is reported at node 2. Four cases are demonstrated: (i) Bernoulli-Euler, $GAK \simeq \infty$, $\rho I = 0$; (ii) Rayleigh, $GAK \simeq \infty$, $\rho I = 2.5 \times 10^{-5}$ kg-m; (iii) Timoshenko (shear only), $GAK = 3.825$ MN, $\rho I = 0$; and (iv) Timoshenko, $GAK = 3.825$ MN, $\rho I = 2.5 \times 10^{-5}$ kg-m.

The responses for all the above cases are shown in Figure 6. Because the pulse is narrow banded, it travels nearly non-dispersively with a group speed of 2.75 km/s. (Note that at

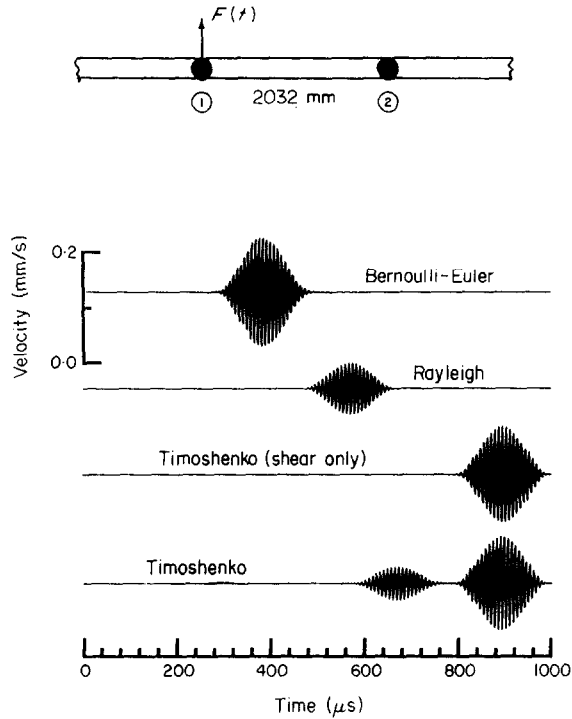


Figure 6. Results for example (a).

initiation the center of the pulse was at 200 μ s.) The first case corresponds to the elementary beam. It is evident from the figure that the waves in the elementary beam travel faster than the waves in the Timoshenko beam. The fourth case is the general Timoshenko beam, and two definite pulses are discernable. They are predicted accurately from the dispersion curves of Figure 2. The second and the third cases correspond to the two intermediate theories. The Rayleigh theory, which accounts for rotational inertia but assumes no shear deformation, predicts a speed similar to the faster Timoshenko mode. The third is when there is only shear deformation. The speed exhibited by it corresponds to the slower Timoshenko mode. This example shows graphically the presence of the second propagating mode. It also shows the ease with which the system can be interrogated using the spectral approach.

5.2. EXAMPLE (B): REFLECTIONS FROM BOUNDARIES

In this example, the ability of the formation to predict reflections is demonstrated for two different boundary conditions.

5.2.1. Single boundary

The smoothed triangular pulse shown in Figure 7 was used as input force. Note from its spectrum that there are no significant amplitudes beyond 70 kHz. Two spectral elements are used here, one of which is a throw-off element as shown in Figure 8(a). All elements are assumed to have the same properties. The beam is impacted at 254 mm from the boundary which is taken as being pinned and the response is also measured at the impact site. The time reconstruction is also shown in Figure 8(a). We can see the difference in response between the Bernoulli-Euler beam and the Timoshenko beam only in the early

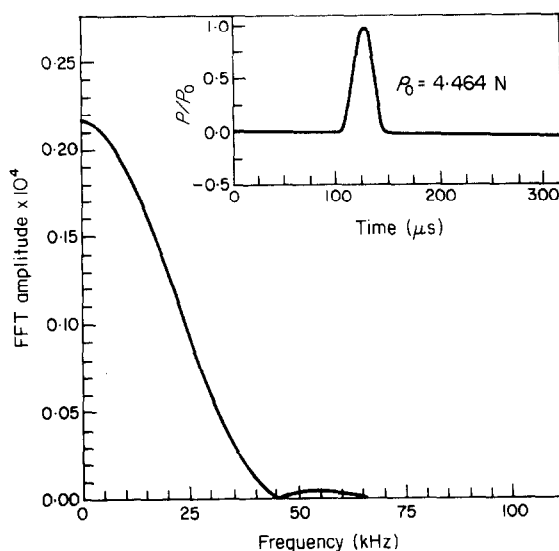


Figure 7. Input force for example (b).

arrival time behavior of the reflected waves. The initial behavior at the impact site is essentially the same for both models. Thus the presence of reflections are implied in the modeling and they will automatically manifest themselves depending on the boundary conditions.

5.2.2. All Finite Boundaries

Here the comparison of results between the spectral approach and the conventional finite elements is provided. There is an interesting contrast between the conventional and the spectral formulations. The former are ideal for solving problems over a short time and where the structure is small in extent; the latter is ideal when the structure is large in extent. Just as the conventional method has difficulty with a semi-infinite boundary, the spectral method has difficulty when all the boundaries are finite. The origin of the problem lies in the fact that since reflections are implicit in the spectral modeling, the bounded problems will automatically exhibit resonance (or sharp spectral peaks) in the frequency domain. To reconstruct these in the time domain would take an enormously large time window (to allow the oscillations to die down and not wrap around). The following example of a cantilever beam (with two finite boundaries) demonstrates how easily time domain reconstructions can nonetheless be obtained with the spectral approach.

The finite element analysis was performed using the program STRIDYN [9]. About 5000 2-D plane stress triangular elements were used to model a cantilever beam of length 508 mm. The spectral model consists of one spectral element of length 508 mm and a throw-off element. The elastic and shear modulus of the throw-off element are made very large (but not infinite) in order to simulate the fixed boundary at node 2. The purpose of the throw-off element is to allow some leakage of energy from the cantilever system: thereby a smaller time window is needed for the reconstructions. The configuration is shown in Figure 8(b). The beam is impacted at the tip using the same triangular pulse of the previous case and the response is measured at 254 mm from the tip. The comparison of results, also shown in Figure 8(b), indicates very good agreement between the finite elements and spectral approach. The Bernoulli-Euler beam response is also shown in the

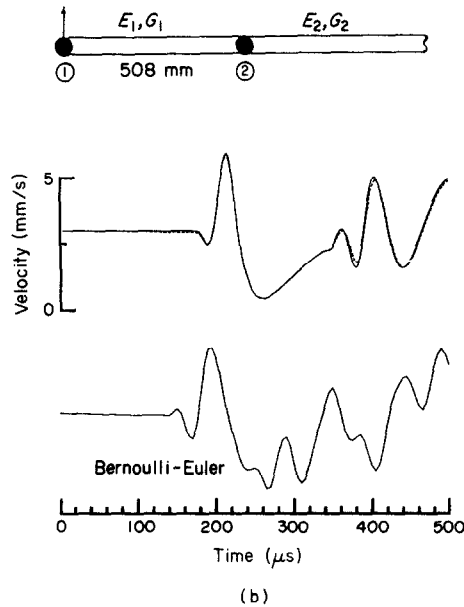
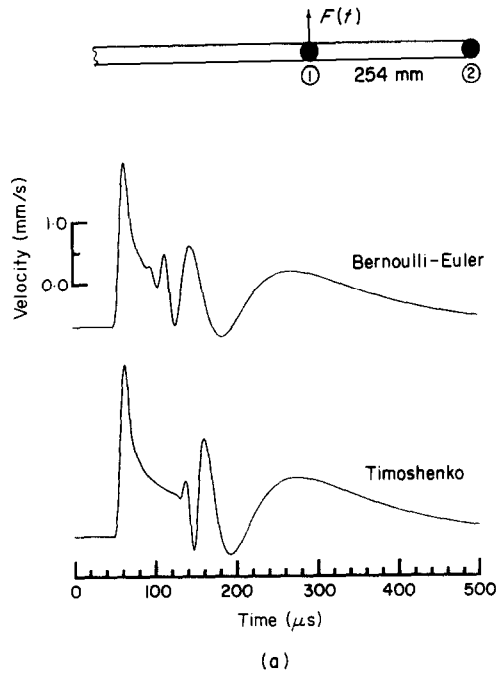


Figure 8. Reflection from (a) pinned boundary and (b) fixed boundary (— STRIDYN, - - -, Timoshenko).

same figure. We can clearly see the difference in the behavior of the two beam models as well as the fact that the reflections are being predicted accurately.

5.3. EXAMPLE (C): MULTIPLE MEMBERS

This final example shows the convenience with which the solution to problems with multiple members can be obtained. Six elements (seven nodes) in total are used, as shown

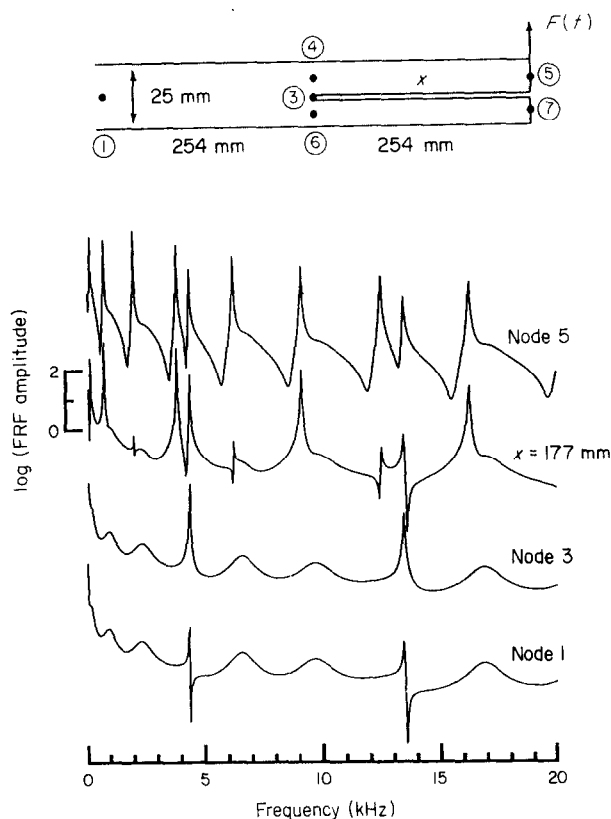


Figure 9. Frequency response function at a number of nodes.

in Figure 9. To make this example a little more interesting without being overly complicated, the original beam is taken as being split along a length of 254 mm. The load is applied in such a way that the problem resembles that of a struck tuning fork. The amplitude of the system response functions are also shown in Figure 9 for nodes 1, 3, 5 and an intermediate point half-way between nodes 4 and 5. The system as a whole exhibits spectral peaks indicating a (damped) resonant type behavior. This is due to the waves reflecting back and forth between nodes 4 and 5 (and 6 and 7).

This example shows that the spectral approach, unlike the conventional finite element method, gives the frequency response of the whole system in a simple and straightforward manner.

6. DISCUSSION

A methodology, analogous to that of the finite element method, is presented that allows problems involving many connected Timoshenko beams to be handled in a convenient and straightforward manner.

Unlike conventional finite elements, however, the length of the spectral element is not a limiting factor; each element is formulated exactly irrespective of its length. Hence, structural connections and discontinuities are the factors which govern the length of the element. This leads to a substantial reduction in the number of equations that are to be solved. The present formulation, which uses a higher order beam theory, does not require

any additional degrees of freedom as compared to the elementary beam formulation; nor does it require additional nodes as for the conventional element.

To complete the split beam analysis referred to at the beginning of the paper, it is necessary to also incorporate the longitudinal motion. This could be done simply using the elementary rod theory to form a frame element (as is already done in the program SPECODYN). However, the opportunity to investigate the consequences of a higher order rod theory will be taken, and this will be the subject of a subsequent paper.

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APPENDIX

Defining the non-dimensional wavenumbers as

$$\xi_1 \equiv k_1 L, \quad \xi_2 \equiv k_2 L,$$

the amplitude ratios are written as

$$P_1 = \frac{iL}{GAK\xi_1} \left(\frac{EI}{L^2} \xi_1^2 + GAK - \rho I \omega^2 \right), \quad P_2 = \frac{iL}{GAK\xi_2} \left(\frac{EI}{L^2} \xi_2^2 + GAK - \rho I \omega^2 \right).$$

Individual terms in matrix $[\hat{Q}]$ referenced in equation (15) are the following:

$$\begin{aligned} \hat{Q}_{11} &= +(r_1 + r_2 e^{-i\xi_2})/\det, & \hat{Q}_{12} &= -P_2(r_1 - r_2 e^{-i\xi_2})/\det, \\ \hat{Q}_{13} &= -(r_1 e^{-i\xi_2} + r_2)/\det, & \hat{Q}_{14} &= P_2(r_1 e^{-i\xi_2} - r_2)/\det, \\ \hat{Q}_{21} &= -(r_1 + r_2 e^{-i\xi_1})/\det, & \hat{Q}_{22} &= P_1(r_1 - r_2 e^{-i\xi_1})/\det, \\ \hat{Q}_{23} &= +(r_1 e^{-i\xi_1} + r_2)/\det, & \hat{Q}_{24} &= -P_1(r_1 e^{-i\xi_1} - r_2)/\det, \\ \hat{Q}_{31} &= -\hat{Q}_{13}, & \hat{Q}_{32} &= \hat{Q}_{14}, & \hat{Q}_{33} &= -\hat{Q}_{11}, & \hat{Q}_{34} &= \hat{Q}_{12}, \\ \hat{Q}_{41} &= -\hat{Q}_{23}, & \hat{Q}_{42} &= \hat{Q}_{24}, & \hat{Q}_{43} &= -\hat{Q}_{21}, & \hat{Q}_{44} &= \hat{Q}_{22}, \end{aligned}$$

where

$$\begin{aligned} r_1 &= (P_1 - P_2)z_{11}, & r_2 &= (P_1 + P_2)z_{12}, & \det &= r_1^2 - r_2^2, \\ z_{11} &= 1 - e^{-i\xi_1} e^{-i\xi_2}, & z_{12} &= e^{-i\xi_1} - e^{-i\xi_2}, & z_{21} &= e^{-i\xi_1} + e^{-i\xi_2}, & z_{22} &= 1 + e^{-i\xi_1} e^{-i\xi_2}. \end{aligned}$$

The following are the stiffness terms in the element stiffness matrix referenced in equation (16):

$$\begin{aligned}
 \hat{k}_{11} &= (\xi_2^2 - \xi_1^2)(r_1 z_{22} + r_2 z_{21})L/det, & \hat{k}_{12} &= [-i\xi_2(r_1 z_{11} + r_2 z_{12}) + i\xi_1(r_1 z_{11} - r_2 z_{12})]L^2/det, \\
 \hat{k}_{13} &= (\xi_1^2 - \xi_2^2)(r_1 z_{21} + r_2 z_{22})L/det, & \hat{k}_{14} &= [-i\xi_1(r_1 z_{12} - r_2 z_{11}) - i\xi_2(r_1 z_{12} + r_2 z_{11})]L^2/det, \\
 \hat{k}_{22} &= (-i\xi_1 P_2 + i\xi_2 P_1)(r_1 z_{22} - r_2 z_{21})L^2/det, & \hat{k}_{24} &= (i\xi_1 P_2 - i\xi_2 P_1)(r_1 z_{21} - r_2 z_{22})L^2/det, \\
 \hat{k}_{23} &= -\hat{k}_{14}, & \hat{k}_{33} &= \hat{k}_{11}, & \hat{k}_{34} &= -\hat{k}_{12}, & \hat{k}_{44} &= \hat{k}_{22}.
 \end{aligned}$$